UNIVERSITY OF COPENHAGEN
FACULTY OF SOCIAL SCIENCES



Master's thesis

Urban Accessibility and Airbnb Prices

Euclidean and walking measures across three Nordic capitals

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Submission date to be confirmed

Abstract

Draft task: Write the final abstract last: problem, three-city data, matched measurements, geographic evaluation, main qualified findings and contribution. Report asking prices and observational evidence. Add verified numbers only.

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1 Introduction

1.1 Motivation and research problem

This thesis examines how straight-line and walking-network representations of urban accessibility relate to Airbnb nightly asking prices in Copenhagen–Frederiksberg, Oslo and Stockholm. It compares what the two representations measure, the information they add to price prediction, geographic variation within cities, and transfer between cities.

Draft task: Explain why accessibility measurement matters for the research problem. Define asking price early. Support the motivation with literature you have read and verified; distinguish listings from bookings or revenue.

1.2 Research gap and contribution

Draft task: Build one argument from the literature to the gap: representation, geographic evaluation and transfer. State the contribution of their combined comparison without claiming universal novelty or a causal effect.

1.3 Research questions

The analysis is organized around four research questions:

RQ1. How do matched Euclidean and walking accessibility measurements differ across the three cities?

RQ2. What information do these measurements add to within-city price prediction, and what complementary adjusted associations are observed?

RQ3. How do predictive performance and accessibility contributions vary across geographic validation designs and held-out neighbourhoods?

RQ4. How do relative-price predictions and accessibility contributions transfer across the six source–target city directions?

1.4 Scope and thesis roadmap

The study uses June 2026 Inside Airbnb listings for conventional entire dwellings. Its evidence is observational. Chapter 2 develops the conceptual framework; Chapter 3 describes the data and

study areas; and Chapter 4 defines measurement and evaluation. Chapter 5 presents the evidence in research-question order, followed by discussion in Chapter 6 and conclusions in Chapter 7.

2 Conceptual Framework and Related Literature

2.1 Accessibility and the spatial context of asking prices

Draft task: Define accessibility, proximity and reachable opportunities. Explain plausible relationships to asking prices, then identify confounding and selection issues that prevent a causal interpretation here.

2.2 Euclidean and walking representations

Draft task: Explain straight-line distance, network restrictions and threshold-based opportunities. Build an expectation for measurement differences without assuming that a richer network measure predicts better.

2.3 Prediction, geographic generalization and transfer

Draft task: Connect price-modelling literature to spatial dependence, geographic holdouts and source–target distribution differences. Distinguish prediction, adjusted association and transfer as separate questions.

2.4 Synthesis and analytical expectations

Draft task: End with a short bridge to RQ1–RQ4. Label expectations as conceptual; do not retrospectively call observed patterns preregistered hypotheses. Avoid another full summary of the introduction.

3 Study Areas and Data

3.1 Study areas and operational neighbourhoods

Draft task: Introduce Copenhagen–Frederiksberg, Oslo and Stockholm. Separate accepted study masks, provider validation groups and native density areas. Explain the 11/17/11 operational groups without certifying them as current official boundaries. Insert study_areas.png and retain its CRS/source notes.

3.2 Sources, dates and geographic coverage

Draft task: Summarize Inside Airbnb, historical OSM networks/destinations and native official population density. Separate acquisition dates from reference dates. State coverage, licensing and vintage limitations. Income and residential prices are excluded. Put detailed provenance in the appendix.

3.3 Sample construction and missingness

Draft task: Walk from raw listings to eligible positive-price references, common predictive samples and genuine complete-case association samples. Use the generated sample-flow and city-summary tables. Explain missing-price exclusion and training-only predictive preprocessing separately.

3.4 Outcome and control variables

Within-city models use the natural logarithm of original observed positive nightly asking prices in native currency. Cross-city transfer uses the city-relative outcome defined in Section 4.6. Missing outcomes are never imputed.

Draft task: Describe the common dwelling, host, amenity, density and centrality controls with units and transformations. Record the native density denominators and vintage caveats, especially Stockholm. Place the complete variable dictionary in Appendix A.

4 Methods

4.1 Analytical design and model comparisons

Let X denote the common controls, E Euclidean accessibility and W walking accessibility. The four model specifications are

$$M_0 = X, \qquad M_E = X + E, \qquad M_W = X + W, \qquad M_{EW} = X + E + W.$$

Within each city, all four predictive models use the same observations, outcomes and evaluation folds. OLS and XGBoost are the specified estimators.

Draft task: Explain the baseline, accessibility-versus-baseline comparisons, walking replacement ($M_W - M_E$) and walking add-on ($M_{EW} - M_E$). Introduce a compact design diagram only if it makes the sequence clearer.

4.2 Constructing matched accessibility measurements

Draft task: Define nearest rail/bus access and food/social and culture/tourism opportunities on matched category universes. Explain the historical OSM timestamp, directed graph and both snap connectors. The primary comparison uses 5 km/h, 10 minutes and a rounded matched 833 m radius. Nearest destinations may differ between representations; do not label their excess same-object route circuity. Distinguish measured zeros from routing/source missingness.

4.3 Geographic validation and training procedure

Draft task: Explain leave-one-operational-neighbourhood-out evaluation and the fixed 15-test/500-training support gate. Then describe random five-fold and fixed-grid comparisons. Explain training-only fitted preprocessing and the 30-candidate, four-inner-spatial-fold XGBoost search. Outer test folds never select variables, thresholds, preprocessing or hyperparameters.

4.4 Metrics, paired contrasts and stability

Draft task: Define RMSE and supporting metrics on their outcome scales. Define candidate-minus-reference contrasts so negative RMSE change means improvement. Explain 2,000 group resamples as conditional geographic-composition stability, with overlapping training data and few groups. These ranges do not establish independent-sample inference or equivalence.

4.5 Complementary complete-case associations

Draft task: Separate adjusted OLS associations from predictive evaluation. Explain common complete cases on $X + E + W$, HC3, restricted Webb inference and spatial diagnostics/sensitivities. Disclose the uniform 5% to 10% eligibility amendment on 13 October 2026 after observing Stockholm's missingness. Missing bedrooms and outcomes remain unfilled in association inference.

4.6 Source-only cross-city transfer

For listing i in city c, the transfer outcome is

$$y_{ic}^{\text{rel}} = \log(p_{ic}) - \mu_c,$$

where p_{ic} is observed positive nightly asking price and μ_c is the frozen mean log price of the city's eligible inside-mask positive-price reference population before feature filtering.

Draft task: Describe all six directions. Preprocessing, tuning and fitting use the source city only. Target prices supply the frozen target benchmark and evaluation, with no target refitting or recalibration. Explain why this is a relative-price transfer test and why directions are dependent.

4.7 Registered robustness and reproducibility

Draft task: Group the completed checks by validation, sample/control, accessibility and association purpose. Explain what each challenges rather than repeating every setting. Keep full configurations and NOT_APPLICABLE statuses in Appendix C. Record the metric CRSs: EPSG:25832 (Copenhagen), EPSG:25833 (Oslo), EPSG:3006 (Stockholm). Describe code/SQL provenance without exposing private records.

5 Results

5.1 RQ1: Accessibility measurement differences

Draft task: Start with a direct answer. Compare nearest-category access and opportunities with units and city differences. Insert the generated summaries, then close with what this establishes for the modelling comparison.

5.2 RQ2: Within-city prediction and adjusted associations

Predictive performance and accessibility contributions

Draft task: Show all four models and both estimators before emphasizing a contrast. Separate baseline gains, replacement and add-on. The frozen XGBoost point gains in Copenhagen/Stockholm and adverse Oslo changes require city-specific qualifications and stability ranges.

Complementary adjusted associations

Draft task: Summarize the complete-case association evidence with units and uncertainty. Keep full coefficients/diagnostics in the appendix. Do not equate coefficients or descriptive feature contributions with causal effects or held-out predictive gains.

5.3 RQ3: Geographic generalization and heterogeneity

Draft task: Compare primary, random and fixed-block tasks, then show neighbourhood variation. Explain reversals and unequal support. Summarize registered robustness patterns without ranking scores across changed samples or counting overlapping grids as independent replications.

5.4 RQ4: Directional cross-city transfer

Draft task: Cover every direction and both estimators. Explain relative-price errors, baseline gains, walking replacement/add-on and directional asymmetry. Do not average away reversals or imply six independent replications.

5.5 Summary of answers

Draft task: Give one concise, evidence-qualified answer to each question. Use this as the bridge to discussion; do not repeat all tables or numerical results. Check every statement against the current claims document.

6 Discussion

6.1 What accessibility representation changes

Draft task: Connect RQ1 measurement differences to RQ2 predictive patterns and the literature. Explain why representational detail and predictive gain can diverge. Distinguish observed evidence from plausible explanations.

6.2 Geography and the limits of transportability

Draft task: Read RQ3 and RQ4 together. Discuss validation geography, source–target differences and directional dependence. Explain the scope of generalization supported by three observed city samples.

6.3 Limitations and methodological decisions

Draft task: Connect each limitation to its consequence: asking-price and missing-outcome selection, displaced listing coordinates, OSM completeness, native density/vintage mismatches, operational groups, few clusters, correlated features and residual spatial dependence. Disclose decision timing and the post-observation 10% amendment. Transfer also requires observed target prices for centring. Avoid unsupported causal housing/displacement claims.

6.4 Implications and future research

Draft task: State what a researcher learns about choosing accessibility representations and evaluation geography. Tie future work to specific limitations. No-density prediction, same-object circuity and median-centred transfer are possible future analyses, not completed study results.

7 Conclusion

Draft task: Answer RQ1–RQ4 briefly, then state the combined contribution and its scope. End on what the comparison establishes about accessibility, geographic evaluation and transfer. Do not add new results or a universal walking-superiority claim.

A Data Provenance and Variable Definitions

Draft task: Include source/reference/acquisition dates, geographic extents, CRSs, checksums where available, sample-flow details, category definitions, controls and qualified native density denominators. Use aggregate evidence.

B Extended Results and Diagnostics

Draft task: Place full associations, descriptive held-out contributions, neighbourhood detail, transfer diagnostics and spatial residual checks here. Retain source captions and privacy suppression for published maps/bins.

C Robustness and Specification Decisions

Draft task: Include the complete generated registry, fixed-grid/settings details and explicit non-applicable entries. Record decision chronology, especially support gates and the association amendment. Historical drafts and unexecuted suggestions are not current results.

References

Draft task: Add checked entries to references.bib and cite them using `\parencite{key}` or `\textcite{key}`. The initial bibliography is intentionally empty; no citations are invented.